A Tutorial on Y-wise Affine Neural Networks:

Neural Representations of Piecewise Affine Functions

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Special Tutorial Session – April 22nd, 2026

Outline

- **Multi-parametric Model Predictive Control (mp- MPC)**
- **Demonstration I**
  - Walk-through applying YANNs
- **Y-wise Affine Neural Networks (YANNs)**
  - Mathematical Foundation
  - Walk-through Examples
- **Demonstration II**
  - Walk-through building a YANN
- **Wrap Up Discussions with Q&A Session**

Multi-Parametric Model Predictive Control

mp-MPC generates explicit control law, which is expressed as affine functions of the defined parametric set (e.g., disturbance, state variables, reference trajectories, etc.)

- Consider the regulation of a discrete-time linear time-invariant system:

$$
\begin{aligned}
\min_{u} \quad & \sum_{k=1}^{N} \left(x_k^T Q x_k + u_k^T R u_k\right) + x_N^T P x_N \\
\text{s.t.} \quad & x_{k+1} = A x_k + B u_k, \quad \forall k \in \{1, 2, \cdots, N\} \\
& y_k = C x_k + D u_k, \quad \forall k \in \{1, 2, \cdots, N\} \\
& x_k \in \mathcal{X}, y_k \in \mathcal{Y}, \quad \forall k \in \{1, 2, \cdots, N\} \\
& u_k \in \mathcal{U}, \quad \forall k \in \{1, 2, \cdots, N\} \\
& x_N \in \mathcal{X}_f
\end{aligned}
$$

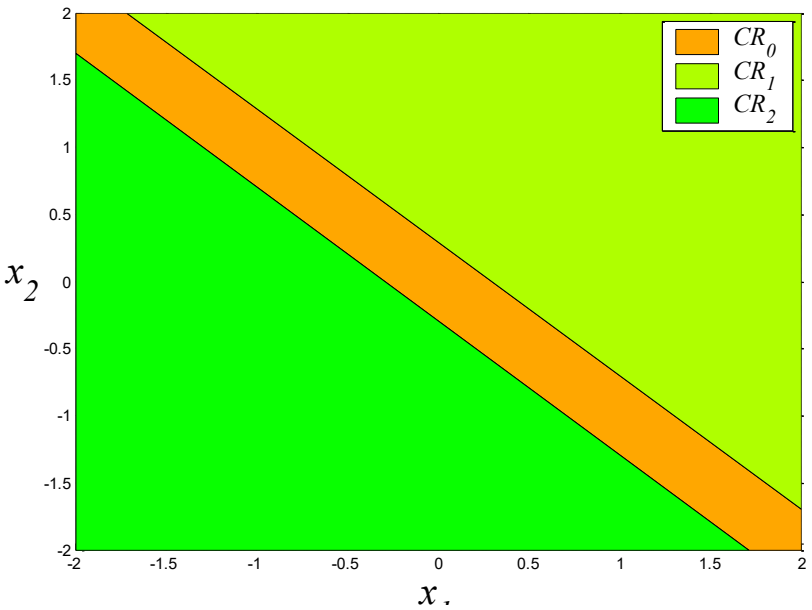
- This can be reformulated into the equivalent mp-QP problem:

$$
\begin{aligned}
J^*(\theta) = \min_{u} \quad & u^T H u + u^T Z \theta + \theta^T \hat{M} \theta \\
\text{s.t.} \quad & G u \leq S \theta + W \\
& C R_A \theta \leq C R_b
\end{aligned}
$$

- The solution to this gives u as a continuous piecewise affine function defined on polytopes (i.e., critical regions)

$$
u^*(\theta) = \begin{cases} K_1 \theta + r_1, \theta \in CR^1 = \{CR_A^1 \theta \leq CR_b^1\} \\ \quad \vdots \\ K_p \theta + r_p, \theta \in CR^p = \{CR_A^p \theta \leq CR_b^p\} \end{cases}
$$

An Example of Explicit Control Laws



$$u_T = K_i\theta_T + r_i \quad \theta_T \in CR^i = \{CR_i^A\theta \le CR_i^b\}$$

$$u(t) = \begin{cases} \begin{bmatrix}-6.8355 & -6.8585\end{bmatrix}x(t) & \text{if } \begin{bmatrix}0.7059 & 0.7083\\ -0.7059 & -0.7083\end{bmatrix}x(t) \le \begin{bmatrix}0.2065\\ 0.2065\end{bmatrix} \\ -2 & \text{if } \begin{bmatrix}-0.7059 & -0.7083\end{bmatrix}x(t) \le -0.2065 \\ 2 & \text{if } \begin{bmatrix}0.7059 & 0.7083\end{bmatrix}x(t) \le -0.2065 \end{cases}$$

$$J(x_{t|t}) = \min_{u_{t|t},\, u_{t+1|t}} \sum_{j=0}^{1}(x_{t+j|t}^T\, x_{t+j|t} + 0.01u_{t+j|t}^2)$$

$$s.t. \quad x_{t+j+1|t} = \begin{bmatrix}0.7326 & -0.0861\\ 0.1722 & 0.9909\end{bmatrix}x_{t+j|t} + \begin{bmatrix}0.0609\\ 0.0064\end{bmatrix}u_{t+j|t}$$

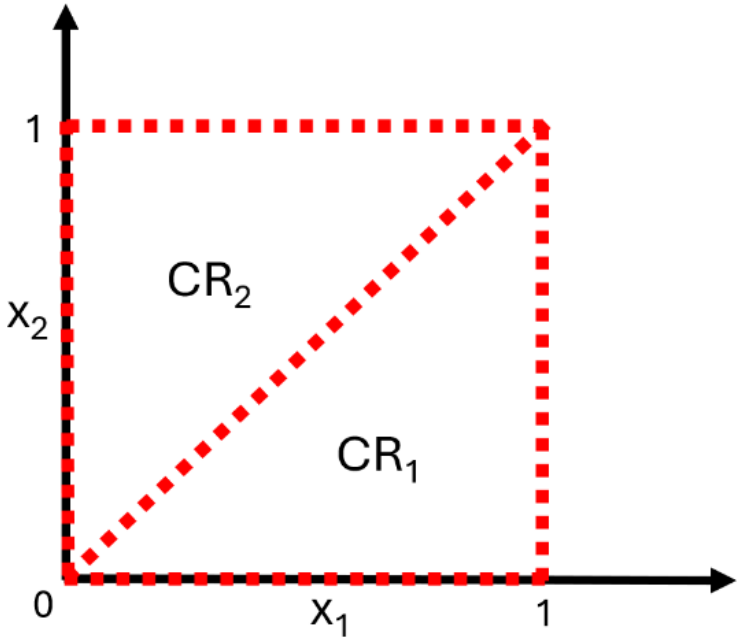
$$-2 \le u_{t+j|t} \le 2, \qquad j = 0,1$$

constraints on the state in the objective function inequalities

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



Question: What should the output be when:

- x=[0.5, 0.4]?
- x=[1, 0]?
- x=[0, 1]?

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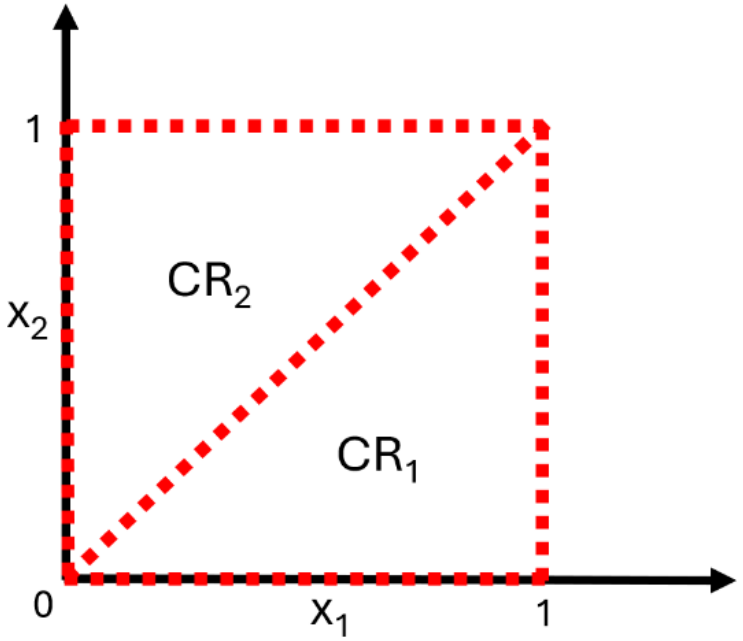


Question: What should the output be when:

- x=[0.5, 0.4]? **f(x) = 0.5**
- x=[1, 0]? **f(x) = 1**
- x=[0, 1]? **f(x) = 1**

Why YANNs?

Case Study 1: Regulation MPC

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} C = \begin{bmatrix} 1 & 2 \end{bmatrix} D = 0$$

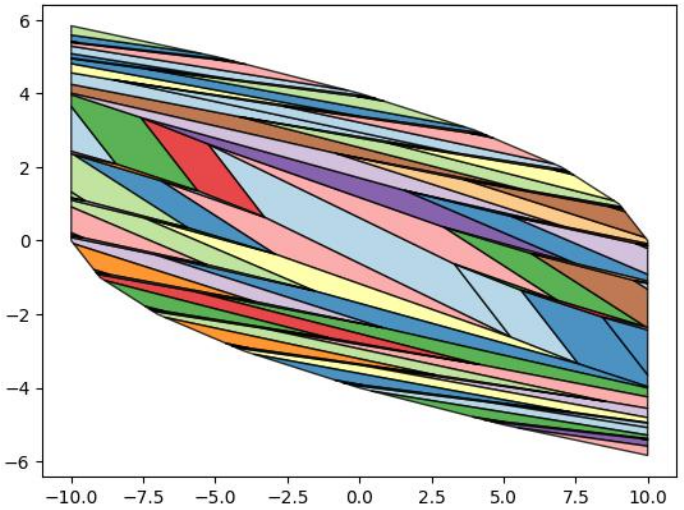
$$-10 \leq x \leq 10$$

$$-1 \leq u \leq 1$$

$$-25 \leq y \leq 25$$

$$Q = I_2,\ R = 0.01,\ N = 10$$

115 critical regions



	mp-MPC	**YANN**
1000 Inferences (s)	0.0223	0.0077
Avg. Time (μs)	22.3	7.7

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Demonstration I

Please go to the following link:

- https://github.com/Innovation-Driven-Sustainable-Systems/yanns

Or follow the QR code:

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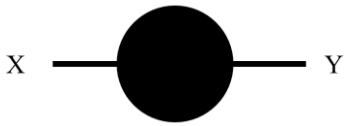
YANN Representation - Constraint Checking

We want to represent the indicator functions for *a single inequality*



$$a_1x_1 + a_2x_2 + \cdots + a_nx_n \leq b$$

$$\mathbb{1}(x) = \begin{cases} 1, & a_1x_1 + a_2x_2 + \cdots + a_nx_n \leq b \\ 0, & \text{else} \end{cases}$$

$$-a_1x_1 - a_2x_2 - \cdots - a_nx_n + b \geq 0$$

$$BSF(x) = \begin{cases} 1, & -a_1x_1 - a_2x_2 - \cdots - a_nx_n + b \geq 0 \\ 0, & else \end{cases}$$

$$W = [-a_1, -a_2, \cdots - a_n] \quad B = b$$

$$\begin{aligned} Y &= \sigma(WX + B) \\ &= \sigma([-a_1, -a_2, \cdots - a_n] \times [x1, x2, \cdots x_n]^T + b) \\ &= \sigma(-a_1x_1 - a_2x_2 - \cdots - a_nx_n + b) \\ &= \begin{cases} 1, & -a_1x_1 - a_2x_2 - \cdots - a_nx_n + b \geq 0 \\ 0, & else \end{cases} \end{aligned}$$

$$\therefore Y = \mathbb{1}(x)$$

Example Problem

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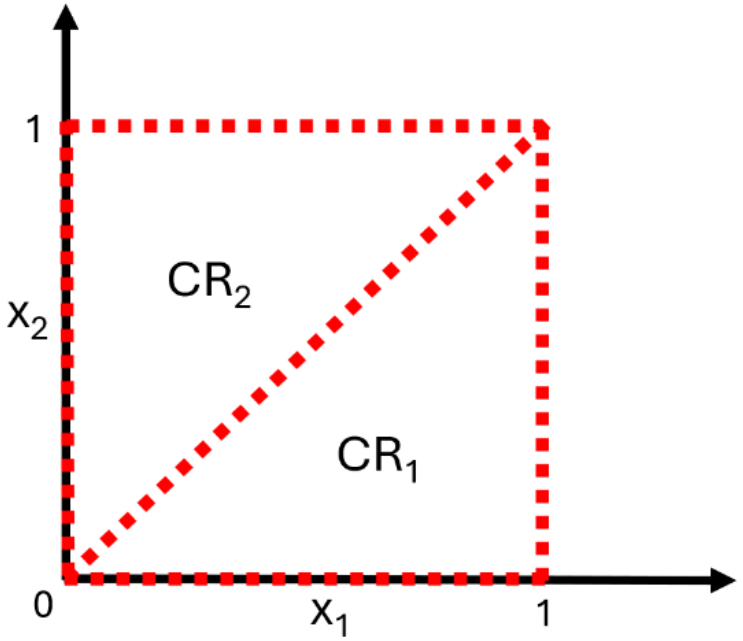
$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



Question: What should the weight and bias matrices be for the circled constraint?

A. W=[1, 1] B = [0]
B. W=[1, -1] B = [0]
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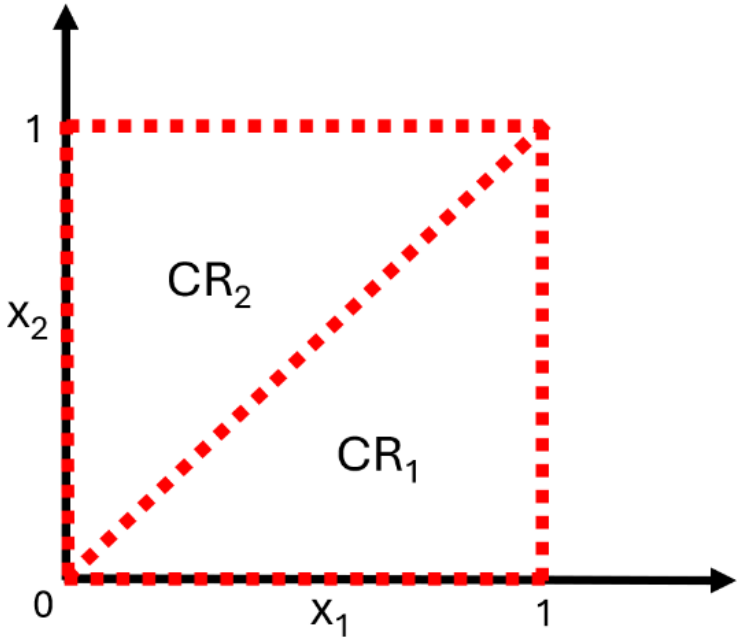

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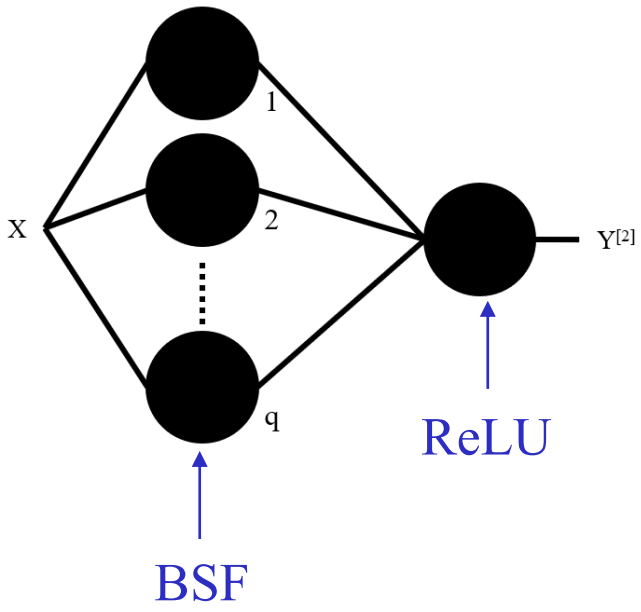
YANN Representation - Constraint Checking

We want to represent the indicator functions for *a system of inequalities*



$$\begin{cases} -a_{1,1}x_1 - a_{2,1}x_2 - \cdots - a_{n,1}x_n + b_1 \geq 0 \\ -a_{1,2}x_1 - a_{2,2}x_2 - \cdots - a_{n,2}x_n + b_2 \geq 0 \\ \vdots \\ -a_{1,q}x_1 - a_{2,q}x_2 - \cdots - a_{n,q}x_n + b_q \geq 0 \end{cases} \qquad \mathbb{1}(x) = \begin{cases} 1, & \begin{cases} \mathbb{1}_1(x) = 1 \\ \mathbb{1}_2(x) = 1 \\ \vdots \\ \mathbb{1}_q(x) = 1 \end{cases} \\ 0, & \text{else} \end{cases}$$

$$W^{[1]} = \begin{bmatrix} -a_{1,1} & -a_{2,1} & \cdots & -a_{n,1} \\ -a_{1,2} & -a_{2,2} & \cdots & -a_{n,2} \\ \vdots & \vdots & \vdots & \vdots \\ -a_{1,q} & -a_{2,q} & \cdots & -a_{n,q} \end{bmatrix}, B^{[1]} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_q \end{bmatrix}$$

$$W^{[2]} = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \in \mathbb{R}^q, B^{[2]} = 1 - q$$

$$\begin{aligned} Y^{[2]} &= \sigma^{[2]}(W^{[2]}Y^{[1]} + B^{[2]}) \\ &= \sigma^{[2]}(\sum_{l=1}^{q} Y^{[1]} - q + 1) \\ &= \max(\sum_{l=1}^{q} Y^{[1]} - q + 1, 0) \\ &= \begin{cases} 1, & \sum_{l=1}^{q} Y^{[1]} = q \\ 0, & \text{else} \end{cases} \end{aligned}$$

$$\therefore Y^{[2]} = \mathbb{1}(x)$$

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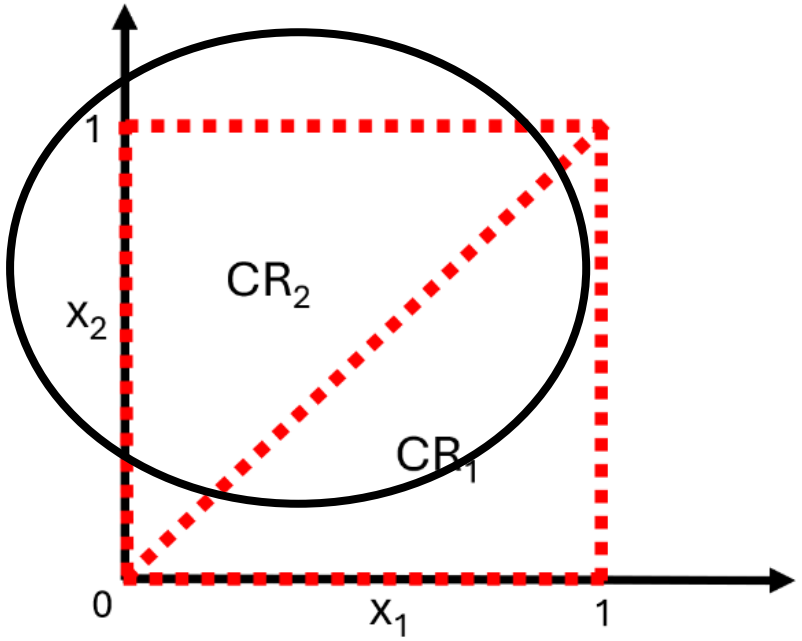

Question: What should the weight and bias matrices be for the 2nd network layer, if we are determining if CR2 is active?

A. W=[1, 1, 1] B = [-1]
B. W=[1, 1] B = [-1]
C. W=[1, 1] B = [-2]
D. W=[1, 1, 1] B=[-2]

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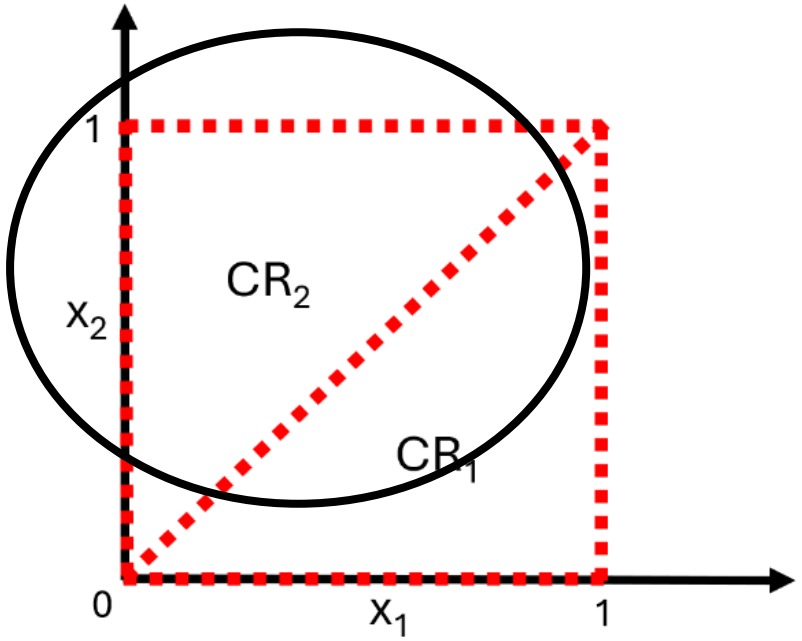

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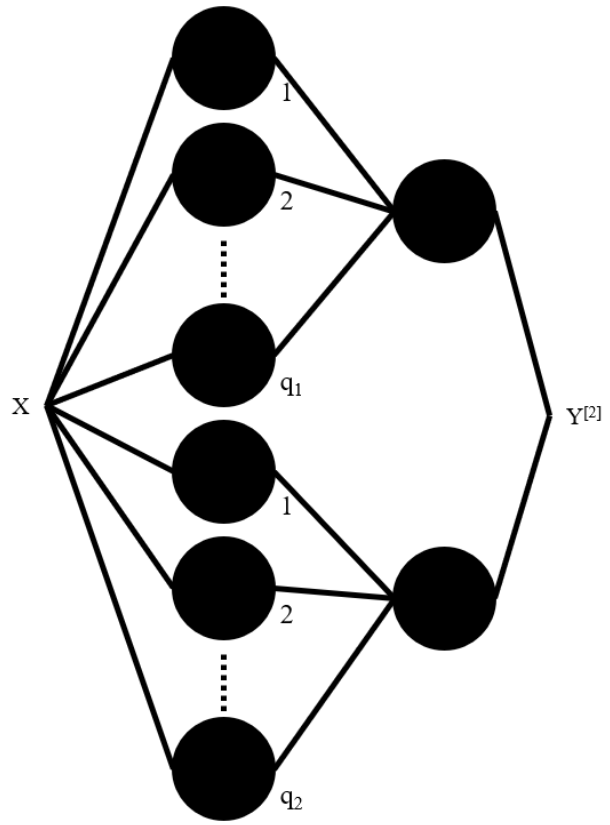
We want to represent the indicator functions for *multiple systems of inequalities*



$$\begin{cases} -a_{1,1,s}x_1 - a_{2,1,s}x_2 - \cdots - a_{n,1,s}x_n + b_{1,s} \geq 0 \\ -a_{1,2,s}x_1 - a_{2,2,s}x_2 - \cdots - a_{n,2,s}x_n + b_{2,s} \geq 0 \\ \vdots \\ -a_{1,q_s,s}x_1 - a_{2,q_s,s}x_2 - \cdots - a_{n,q_s,s}x_n + b_{q_s,s} \geq 0 \end{cases}$$

$$W^{[1]} = \begin{bmatrix} -a_{1,1,1} & \cdots & -a_{n,1,1} \\ \vdots & \vdots & \vdots \\ -a_{1,q_1,1} & \cdots & -a_{n,q_1,1} \\ -a_{1,1,2} & \cdots & -a_{n,1,2} \\ \vdots & \vdots & \vdots \\ -a_{1,q_p,p} & \cdots & -a_{n,q_p,p} \end{bmatrix}, B^{[1]} = \begin{bmatrix} b_{1,1} \\ \vdots \\ b_{q_1,1} \\ b_{1,2} \\ \vdots \\ b_{q_p,p} \end{bmatrix}$$

$$W^{[2]T} = \begin{bmatrix} [1]_{q_1} & 0 & \cdots & 0 \\ [0]_{q_1} & [1]_{q_2} & \vdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & [1]_{q_p} \end{bmatrix}, B^{[2]} = \begin{bmatrix} 1 - q_1 \\ 1 - q_2 \\ \vdots \\ 1 - q_p \end{bmatrix}$$

$$Y^{[2]} = \begin{bmatrix} \begin{cases} 1, & \sum_{l=1}^{q_1} Y_l^{[1]} = q_1 \\ 0, & \text{else} \end{cases} \\ \begin{cases} 1, & \sum_{l=q_1+1}^{q_2+q_1} Y_l^{[1]} = q_2 \\ 0, & \text{else} \end{cases} \\ \vdots \\ \begin{cases} 1, & \sum_{l=q-q_p}^{q} Y_l^{[1]} = q_p \\ 0, & \text{else} \end{cases} \end{bmatrix}$$

$$\therefore Y^{[2]} = \begin{bmatrix} \mathbb{1}_1(x) \\ \mathbb{1}_2(x) \\ \vdots \\ \mathbb{1}_p(x) \end{bmatrix}$$

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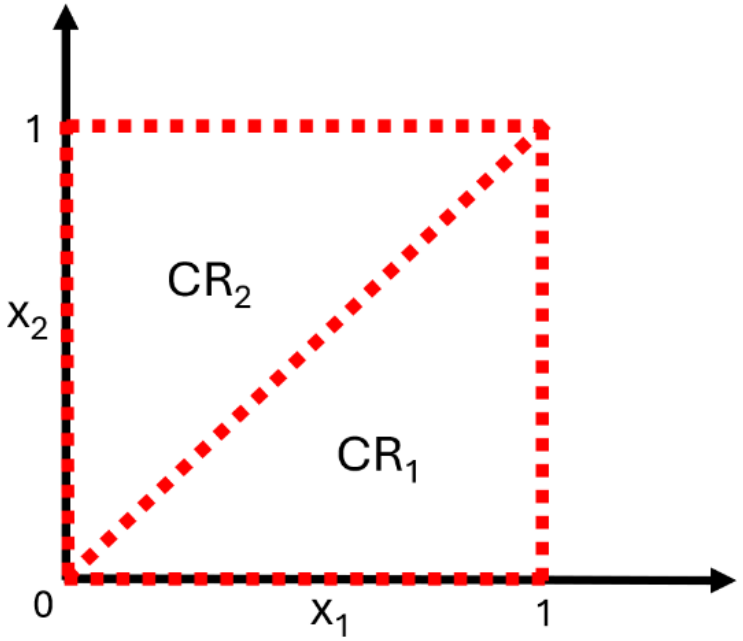

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- x=[1, 0]?
- x=[0, 1]?
- x=[1, 1]?
- x=[0.5, 0.5]?
- x=[2, 2]?

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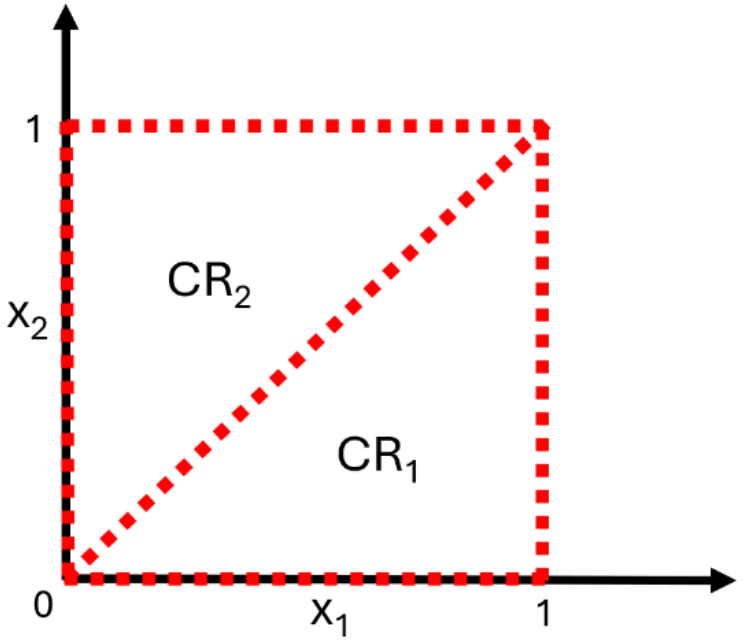

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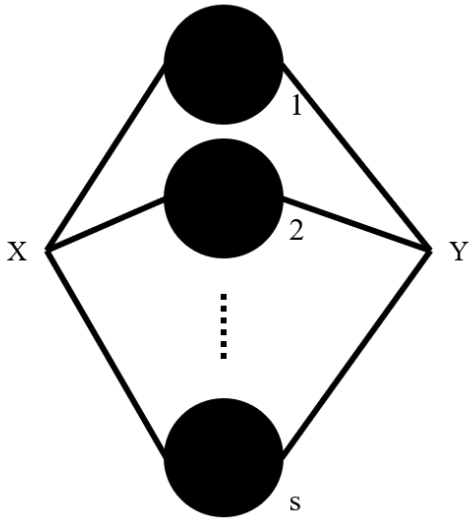
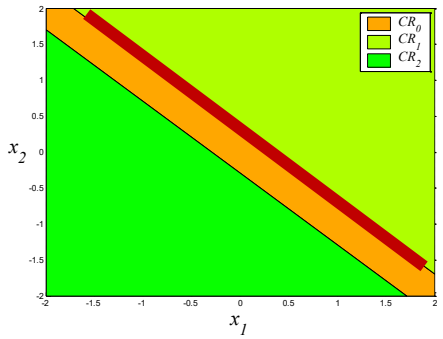
- x=[0.5, 0.5]? **[1, 1]**
- x=[2, 2]? **[0, 0]**

YANN Representation - Critical Region Selection

We want to force *only one critical region to be "active"*





Example:

$$f\left(\begin{bmatrix}0\\1\\1\\0\end{bmatrix}\right) = \begin{bmatrix}0\\1\\0\\0\end{bmatrix}$$

$$W = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & -1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ -1 & -1 & -1 & -1 & \cdots & 1 & 0 \\ -1 & -1 & -1 & -1 & \cdots & -1 & 1 \end{bmatrix}, B = 0$$

$$Y = \begin{bmatrix} \max(d_1, 0) \\ \max(d_2 - d_1, 0) \\ \max(d_3 - (d_2 + d_1), 0) \\ \vdots \\ \max(d_s - \sum_{l=1}^{s-1} d_l, 0) \end{bmatrix}$$

$$Y_i = \max(d_i - \sum_{l=1}^{i-1} d_l, 0)$$

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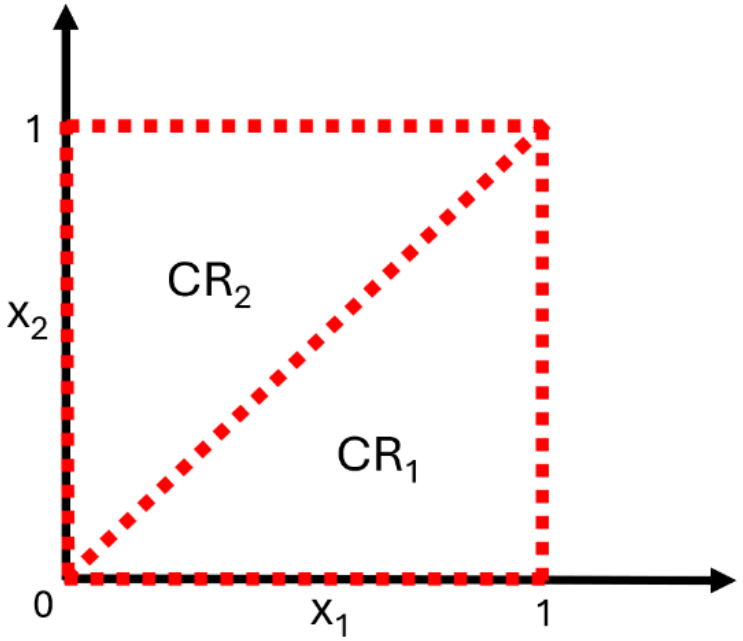

Question: What would be the output of layer 3 if:

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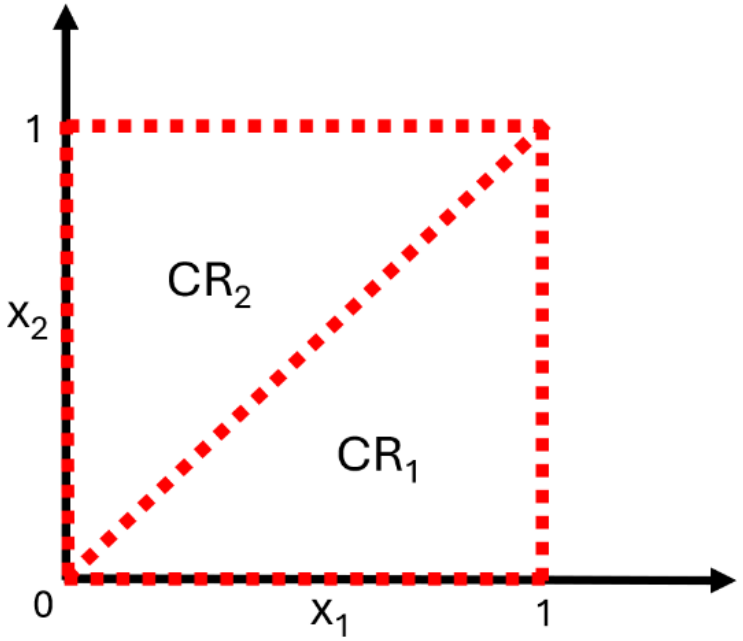

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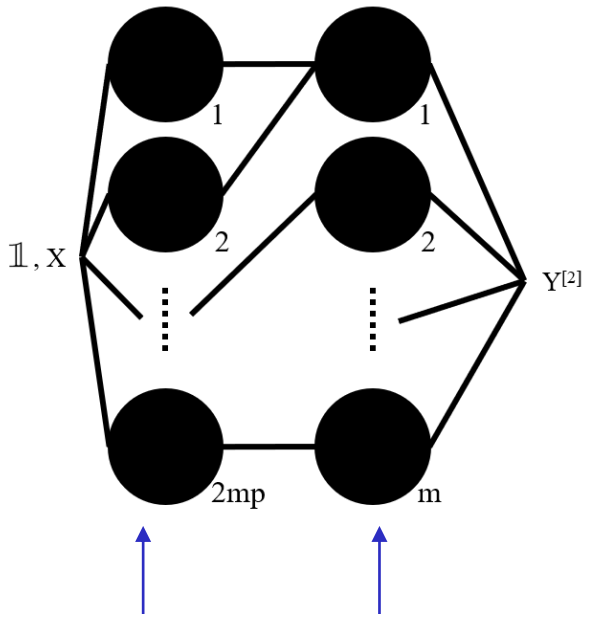
- x=[1, 1]? **[1, 0]**
- x=[0.5, 0.5]? **[1, 0]**
- x=[2, 2]? **[0, 0]**

YANN Representation - Function Evaluation

Given the solution for constraint checking,
we want to *calculate the corresponding control law*



ReLU

linear activation function, $f(x) = x$

$$\begin{cases} a_{1,1,k}x_1 + \cdots + a_{n,1,k}x_n + b_{1,k} = u_{1,k} \\ a_{1,2,k}x_1 + \cdots + a_{n,2,k}x_n + b_{2,k} = u_{2,k} \\ \vdots \\ a_{1,m,k}x_1 + \cdots + a_{n,m,k}x_n + b_{m,k} = u_{m,k} \end{cases}$$

$$W^{[1]T} = \left[\begin{array}{ccccccc} [M]_{2m} & 0 & \cdots & \cdots & \cdots & 0 & 0 \\ [0]_{2m} & [M]_{2m} & \ddots & \cdots & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \cdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & \cdots & \cdots & [M]_{2m} & [0]_{2m} \\ 0 & 0 & \cdots & \cdots & \cdots & 0 & [M]_{2m} \\ \hline a_{1,1,1} & -a_{1,1,1} & \cdots & -a_{1,m,1} & a_{1,1,2} & \cdots & -a_{1,m,p} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n,1,1} & -a_{n,1,1} & \cdots & -a_{n,m,1} & a_{n,1,2} & \cdots & -a_{n,m,p} \end{array}\right]$$

$$B^{[1]} = -M + \begin{bmatrix} b_{1,1} & -b_{1,1} & \cdots & b_{m,1} & -b_{m,1} & \cdots & -b_{m,p} \end{bmatrix}^T$$

$$W^{[2]} = \begin{bmatrix} 1 & -1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & -1 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & -1 \end{bmatrix}_{\times p}, B^{[2]} = 0$$

$$Y^{[2]} = \sigma^{[2]}(W^{[2]}Y^{[1]} + B^{[2]})$$

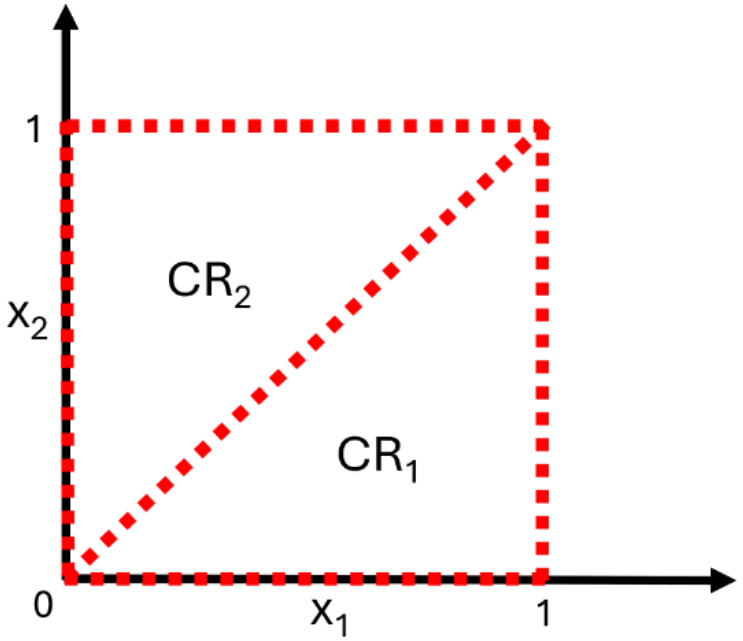
$$= \begin{bmatrix} \sum_{k=1}^{p}(\max(M\mathbb{1}_k(x) + u_{1,k} - M, 0) - \max(M\mathbb{1}_k(x) - u_{1,k} - M, 0)) \\ \vdots \\ \sum_{k=1}^{p}(\max(M\mathbb{1}_k(x) + u_{m,k} - M, 0) - \max(M\mathbb{1}_k(x) - u_{m,k} - M, 0)) \end{bmatrix}$$

$$\therefore Y^{[2]} = f(x) : x \in Dom(f(x)), \sum_{k=1}^{p} \mathbb{1}_k = 1$$

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



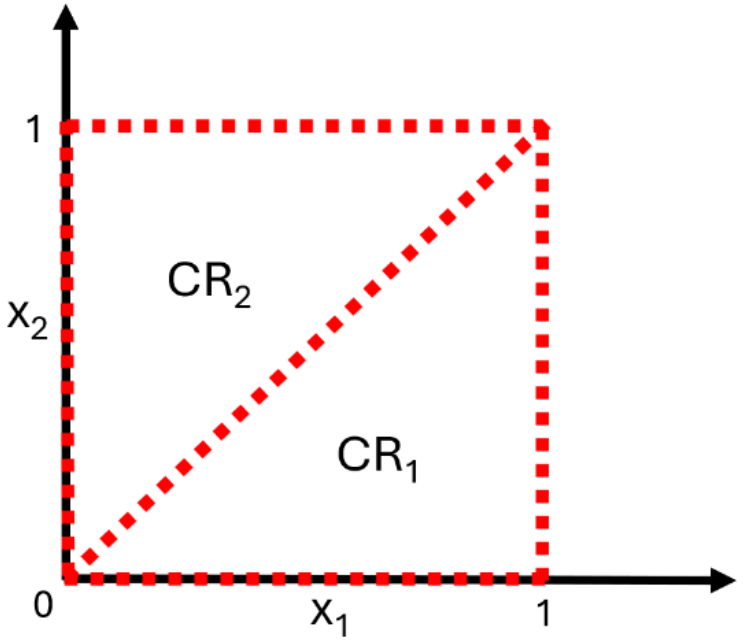
Question: What would be the output of layer 4 if:

- x=[0.5, 0.4]?
- x=[1, 0]?
- x=[0, 1]?
- x=[1, 1]?
- x=[0.5, 0.5]?
- x=[2, 2]?

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



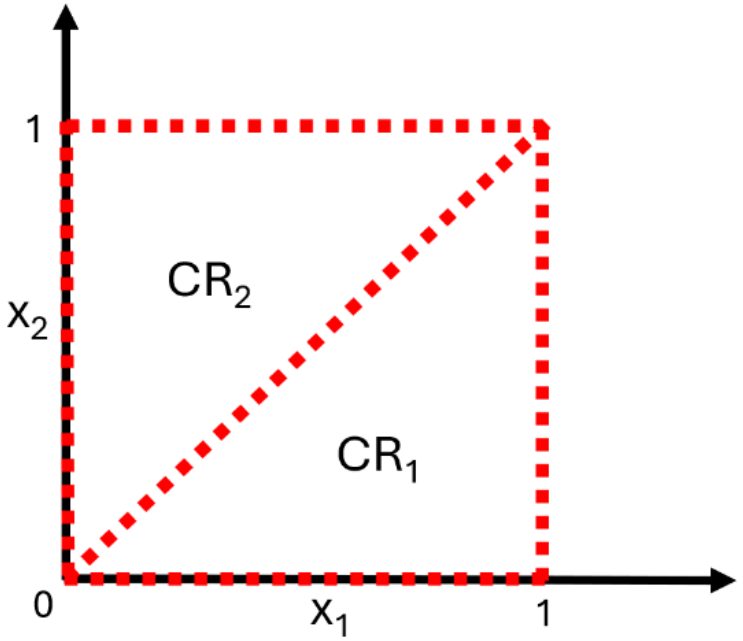
Question: What would be the output of layer 4 if:

- x=[0.5, 0.4]? **[0.5, 0]**
- x=[1, 0]? **[1, 0]**
- x=[0, 1]? **[0, 1]**
- x=[1, 1]? **[1, 0]**
- x=[0.5, 0.5]? **[0.5, 0]**
- x=[2, 2]? **[0, 0]**

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



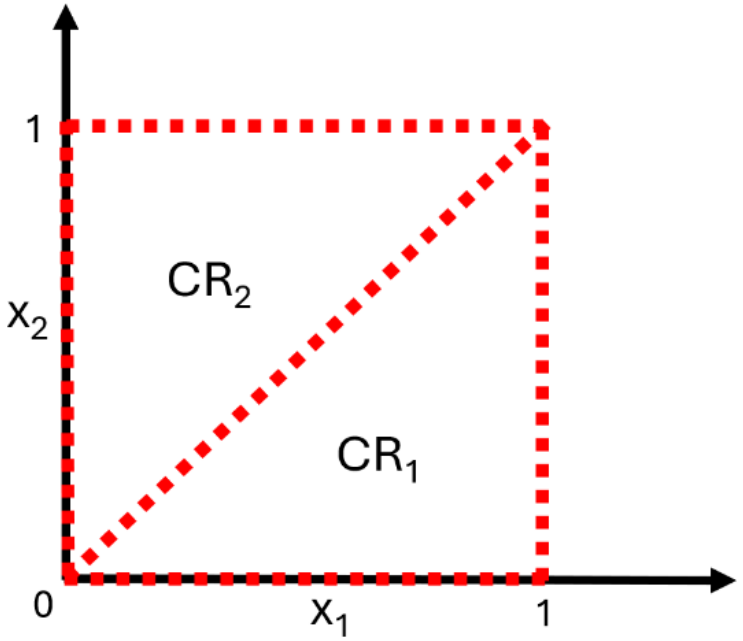
Question: What would be the output of layer 5 if:

- x=[0.5, 0.4]?
- x=[1, 0]?
- x=[0, 1]?
- x=[1, 1]?
- x=[0.5, 0.5]?
- x=[2, 2]?

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



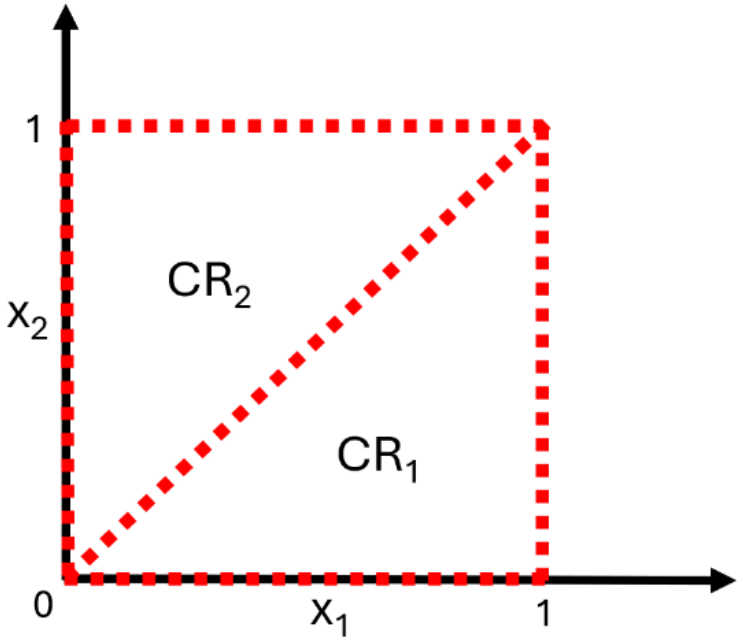
Question: What would be the output of layer 5 if:

- x=[0.5, 0.4]? **[0.5]**
- x=[1, 0]? **[1]**
- x=[0, 1]? **[1]**
- x=[1, 1]? **[1]**
- x=[0.5, 0.5]? **[0.5]**
- x=[2, 2]? **[0]**

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$



Question: Can M take the following values?

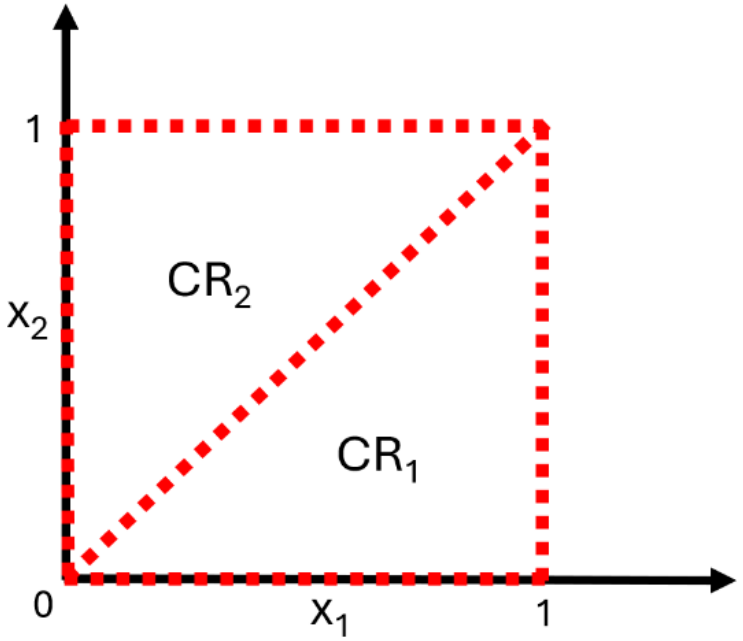
- 0
- -5
- 50
- 5000
- 12e45

Question: What is the smallest value of M that can be assigned?

Example Problem

Consider the following piecewise affine function with the solution map given to the right. Its input has dimension 2 and its output has dimension 1.

$$f(x) = \begin{cases} f_1(x) = x_1, & x \in \{x : 0 \leq x_1 \leq 1, x_1 - x_2 \geq 0\} \\ f_2(x) = x_2, & x \in \{x : 0 \leq x_2 \leq 1, x_1 - x_2 \leq 0\} \end{cases}$$

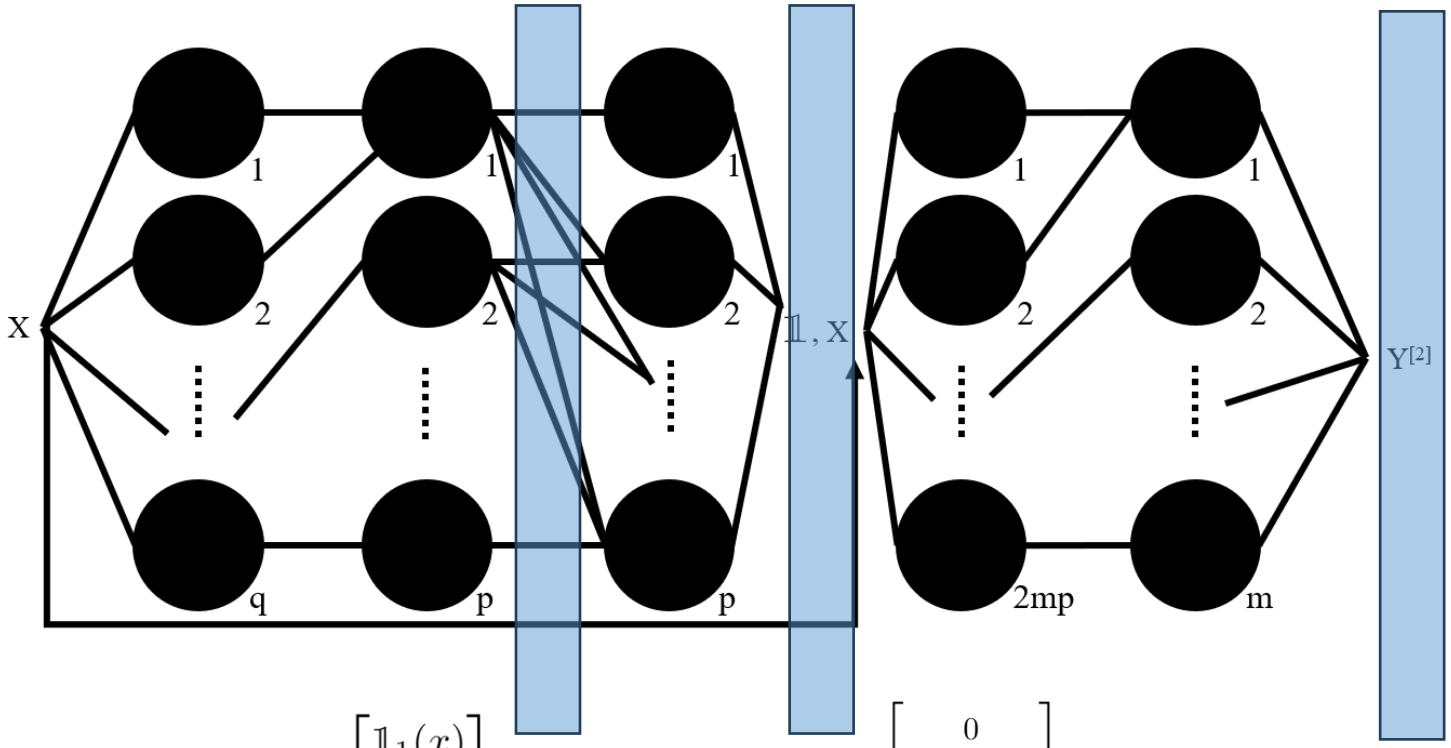


Question: Can M take the following values?

- 0 **no**
- -5 **no**
- 50 **yes**
- 5000 **yes**
- 12e45 **yes (only mathematically…)**

Question: What is the smallest value of M that can be assigned? **M=1**

YANN Representation - Full Network



$$Y^{[2]} = \begin{bmatrix} \mathbb{1}_1(x) \\ \mathbb{1}_2(x) \\ \vdots \\ \mathbb{1}_p(x) \end{bmatrix}$$

$$Y^{[3]} = \begin{bmatrix} 0 \\ \vdots \\ \mathbb{1}_k(x) = 1 \\ \vdots \\ 0 \end{bmatrix}$$

$$Y^{[5]} = f(x) : x \in Dom(f(x))$$

Outline

- **Multi-parametric Model Predictive Control (mp- MPC)**
- **Demonstration I**
  - Walk-through applying YANNs
- **Y-wise Affine Neural Networks (YANNs)**
  - Mathematical Foundation
  - Walk-through Examples
- **Demonstration II**
  - Walk-through building a YANN
- **Wrap Up Discussions with Q&A Session**

Demonstration II

Please go back to the following link:

- https://github.com/Innovation-Driven-Sustainable-Systems/yanns

Or follow the QR code:

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- Dylan Nice
- Dylan Miller
- Ethan Robey

Undergraduate students

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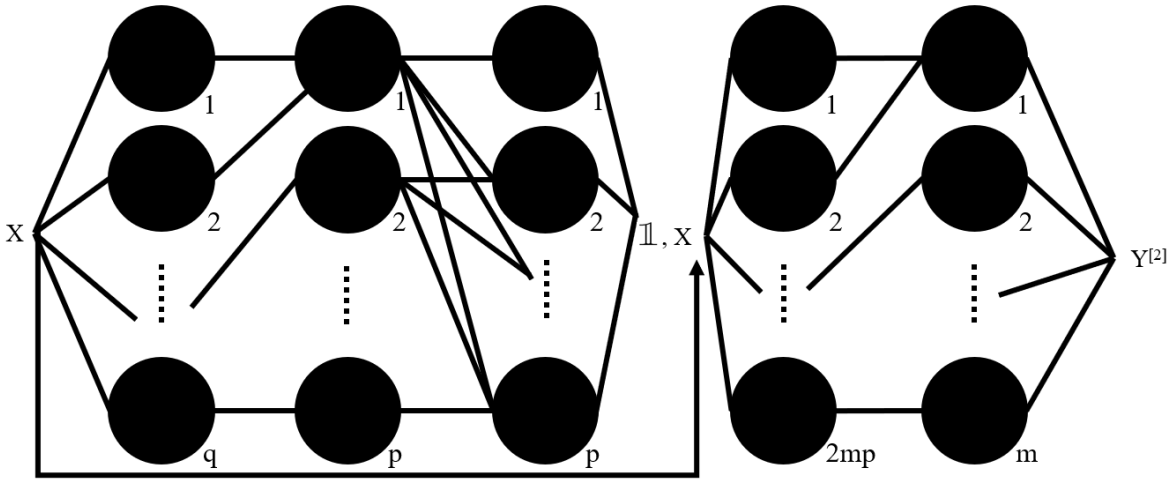
- Ethan Brown
- Nam Khuu
- Jared Marsh
- Adalaide Hickey
- Ethan Metz
- Colton Balog

Wrap Up Discussions and Q&A



YANN Paper – *Computers and Chemical Engineering*

A Tutorial on Y-wise Affine Neural Networks:

Neural Representations of Piecewise Affine Functions

Austin Braniff, Yuhe Tian

Department of Chemical and Biomedical Engineering, West Virginia University,

WV, United States

Special Tutorial Session – April 22nd, 2026